

pitch deck outline (export to PDF for submission)

AI Infra Summit Hackathon - Intel Online track

Title

VoiceSort: language-conditioned bimanual SO-101 manipulation, deployed on

Intel hard

Problem

The Intel challenge: build a MuJoCo scene with SO-101 arms, collect LeRobot

demonstr

Scene and system

- Custom dual-SO-101 MuJoCo scene (namespaced ``<attach>`` composition)
- 3 procedural objects, 2 bins, overhead + wrist cameras
- Voice/text instruction -> MiniLM semantic embedding -> ACT policy token
- Closed-loop 20 Hz control, 12-DoF position actuators

Data

- 115 scripted LeRobot episodes (IK-scripted, physics-executed)
- Designated-arm task table: the instruction determines arm + object + bin
- Honest kinematic-carry data-gen shortcut, documented

- ACT (52M) + `observation.environment\_state` language slot
- Frozen MiniLM embeddings, so rephrased commands work (paraphrase 4/4)
- Instruction-swap video: the wrong instruction reroutes the correct arm

Intel deployment

- ONNX -> OpenVINO IR (normalize + policy + denormalize all in-graph)
- i9-13900K CPU: 28.9 Hz policy rate, 3.2x PyTorch CPU
- UHD 770 iGPU: 28.6 Hz, real iGPU execution rather than a CPU fallback
- Closed-loop success on OV/CPU: 13/24 (matches torch 12/24 within noise)

Voice layer

- Speechmatics STT -> keyword planner -> semantic embedding -> rollout
- Typed-command fallback; declines unparseable commands honestly

Reproducibility and limits

- One-command data gen / train / eval / export / deploy scripts
- Apache-2.0 SO-101 assets + fully procedural objects (zero license risk)
- Limits: grasp-assist abstraction at eval, 3-task scope, ~50% success

ceiling